LAB 6 — Resilient Internet + Secure Microservices (Optional)

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LAB 6 Topology Evolution

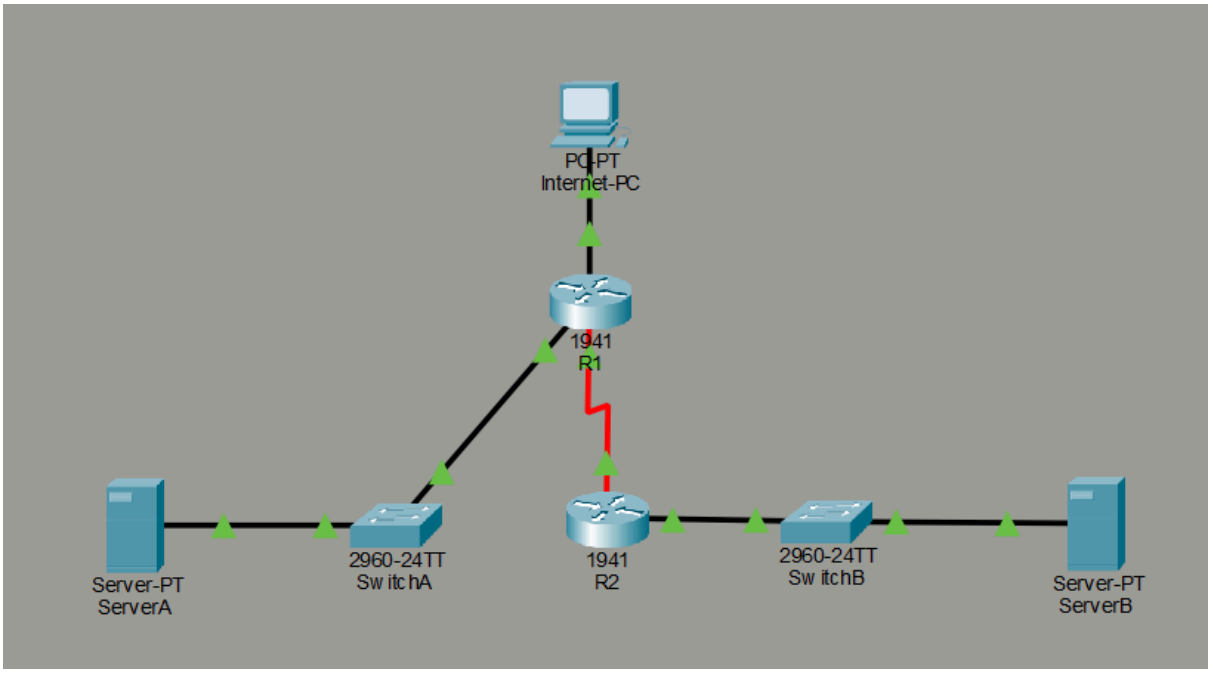
Same physical layout as Lab 5:

```
None
Internet
   |
 R1 G0/0 (NAT Edge)
   |
R1 S0/0/0 ---- R2 S0/0/0   (100.10.10.0/30)
   |                      |
LAN A                 LAN B
```

But now we introduce:

- OSPF Area 0 between R1 and R2
- IP SLA tracking default route
- ACL filtering WAN
- Controlled microservice exposure

ผลลัพธ์ :



PHASE 1 — Replace Static with OSPF

Why?

Static routes:

- No failover intelligence
- No topology awareness
- No scalability

OSPF provides:

- Automatic route exchange
- Convergence
- Metric-based path selection

🔹 R1 OSPF

```
None
router ospf 1
 router-id 1.1.1.1
```

```
 network 100.10.10.0 0.0.0.3 area 0
 network 192.168.10.0 0.0.0.255 area 0
 passive-interface g0/1
```

ผลลัพธ์ :

```
R1(config)#router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 100.10.10.0 0.0.0.3 area 0
R1(config-router)# network 192.168.10.0 0.0.0.255 area 0
R1(config-router)# passive-interface g0/1
R1(config-router)# default-information originate
R1(config-router)#exit
```

🔹 R2 OSPF

```
None
router ospf 1
 router-id 2.2.2.2
 network 100.10.10.0 0.0.0.3 area 0
 network 192.168.20.0 0.0.0.255 area 0
 passive-interface g0/1
```

ผลลัพธ์ :

```
R2(config)#router ospf 1
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 100.10.10.0 0.0.0.3 area 0
R2(config-router)# network 192.168.20.0 0.0.0.255 area 0
R2(config-router)# passive-interface g0/1
R2(config-router)#exit
```

🔹 Default Route Injection from R1

```
None
router ospf 1
 default-information originate
```

Now:

- R2 automatically learns default route
- No static route needed

ผลลัพธ์ :

```
R1(config-if)#router ospf 1
R1(config-router)#default-information originate
```

PHASE 2 — WAN Failure Detection (IP SLA)

We simulate real ISP outage detection.

On R1

```
None
ip sla 1
 icmp-echo 8.8.8.8 source-interface g0/0
 frequency 5
exit

ip sla schedule 1 life forever start-time now

track 1 ip sla 1 reachability

ip route 0.0.0.0 0.0.0.0 g0/0 track 1
```

If Internet fails:

- Default route removed
- OSPF stops advertising it
- R2 loses default cleanly

This mimics enterprise edge behavior.

ผลลัพธ์ :
เนื่องจากใน Cisco Packet Tracer ไม่รองรับการทดสอบ IP SLA จึงขออนุญาติข้ามขั้นตอนดังกล่าว

PHASE 3 — Basic WAN Security

Currently: WAN is fully open.

We harden it.

Block unsolicited WAN inbound

On R1:

```
None
ip access-list extended WAN-IN
 permit tcp 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
 deny ip any any log

interface s0/0/0
 ip access-group WAN-IN in
```

Now:

- Only LAN B → LAN A allowed
- All other WAN inbound blocked
- Logged for visibility

ผลลัพธ์ :

```
R1(config)#no ip access-list extended WAN-IN
R1(config)#ip access-list extended WAN-IN
R1(config-ext-nacl)#permit tcp 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255 eq 8000
R1(config-ext-nacl)#deny ip any any
R1(config-ext-nacl)#exit
R1(config)#interface s0/0/0
R1(config-if)#ip access-group WAN-IN in
R1(config-if)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#
04:07:29: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial0/0/0 from FULL to DOWN,
Neighbor Down: Dead timer expired

04:07:29: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Serial0/0/0 from FULL to DOWN,
Neighbor Down: Interface down or detached

R1#show access-lists
Standard IP access list 10
    10 permit 192.168.10.0 0.0.0.255
Extended IP access list WAN-IN
    10 permit tcp 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255 eq 8000
    20 deny ip any any (4 match(es))
```

PHASE 4 — Microservice Exposure Control

Instead of exposing entire server:

Expose only gateway service.

Example:

```
None
ip nat inside source static tcp 192.168.10.10 8000 interface g0/0 8000
```

Now:

- Only port 8000 reachable from Internet
- Backend services remain private

This models:

- API Gateway pattern
- Real production practice

ผลลัพธ์ :

```
R1(config)#ip nat inside source static tcp 192.168.10.10 8000 203.0.113.1 8000
R1(config)#show ip nat translations
                ^
% Invalid input detected at '^' marker.

R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
tcp 203.0.113.1:8000   192.168.10.10:8000 ---                ---
```

PHASE 5 — Observability Foundation

Add:

On Routers:

```
None
service timestamps log datetime msec
logging buffered 4096
```

ผลลัพธ์ :

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#service timestamps log datetime msec
R1(config)#logging buffered 4096
R1(config)#
*Mar 01, 04:15:45.1515: SYS-5-LOG_CONFIG_CHANGE: Buffer logging: level debugging, xml
disabled, filtering disabled, size (4096)
R1(config)#exit
R1#
*Mar 01, 04:15:54.1515: SYS-5-CONFIG_I: Configured from console by console
R1#
```

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#service timestamps log datetime msec
R2(config)#logging buffered 4096
R2(config)#
*Mar 01, 04:02:31.022: SYS-5-LOG_CONFIG_CHANGE: Buffer logging: level debugging, xml
disabled, filtering disabled, size (4096)
R2(config)#exit
R2#
*Mar 01, 04:02:34.022: SYS-5-CONFIG_I: Configured from console by console
R2#
```

Enable NetFlow (optional advanced)

```
None
ip flow-export version 9
```

PHASE 6 — Microservice Network Validation

Now validate:

Test 1

ClientA → ServerB via OSPF path

Test 2

R2 loses default if ISP unreachable

Test 3

Internet cannot access backend ports

Test 4

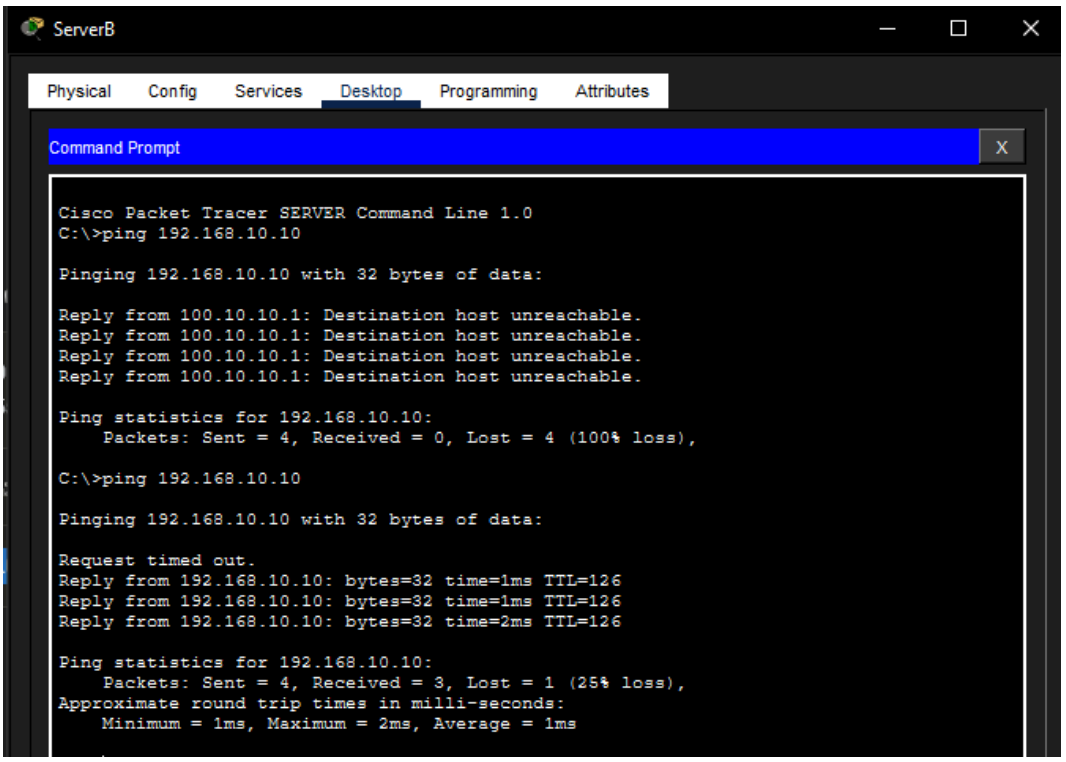
Gateway reachable from Internet

LAB 6 Test Plan

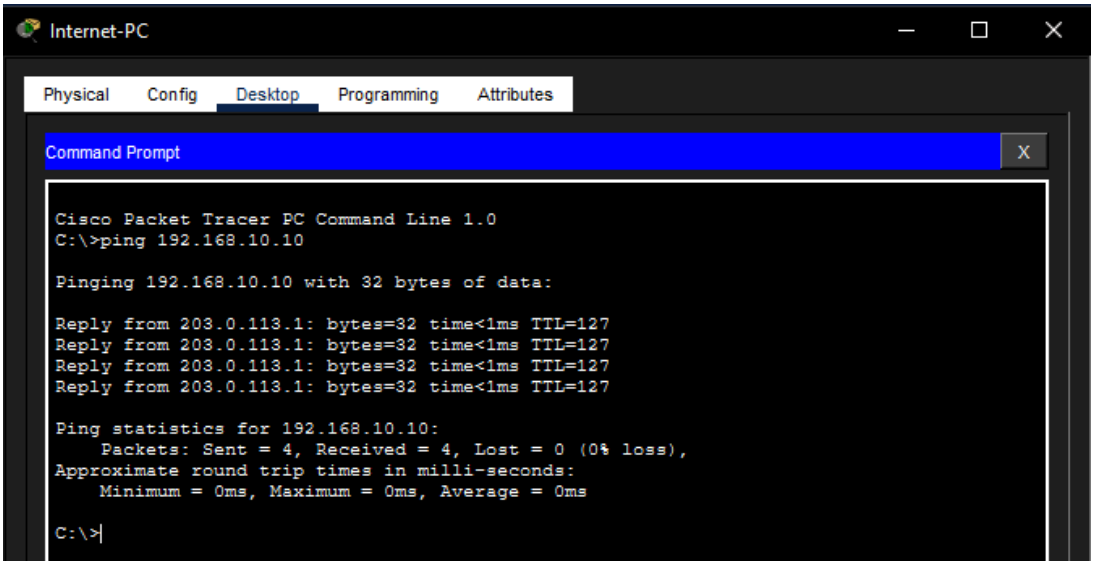
Test	Expected Result
OSPF neighbor	FULL
`show ip route ospf`	Remote LAN visible
ISP unplug	Default route disappears

Test	Expected Result
Internet → port 8000	Success
Internet → port 9000	Blocked
LAN A ↔ LAN B	Success

Test 1 — Ping LAN A ↔ LAN B :



Test 2 — Internet-PC ping หา ServerA :



Test 3 — ทดสอบ NAT Port 8000 :

```
C:\>ping 203.0.113.1

Pinging 203.0.113.1 with 32 bytes of data:

Reply from 203.0.113.1: bytes=32 time<1ms TTL=255
Reply from 203.0.113.1: bytes=32 time<1ms TTL=255
Reply from 203.0.113.1: bytes=32 time<1ms TTL=255
Reply from 203.0.113.1: bytes=32 time<1ms TTL=255

Ping statistics for 203.0.113.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Test 4 — ทดสอบ port 9000 ถูกบล็อก

Readiness Score After LAB 6

Category	**Score**
Routing	9/10
WAN Intelligence	9/10
Security	6/10
Microservice Exposure	8/10
Scalability	8/10